

(Elements of) Categories in Haskell

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Learning outcomes

- DataKinds and GADT for defining categories faithfully
- The module `Control.Category`
- The “category” **Hask**

Recap: What is a Category?

- Objects: abstract entities
- Morphisms: arrows between objects
- Identity morphism: $id_X : X \rightarrow X$
- Composition: $g \circ f : X \rightarrow Z$
- Laws:
 - Identity: $id_Y \circ f = f = f \circ id_X$
 - Associativity: $h \circ (g \circ f) = (h \circ g) \circ f$

Categories in Haskell with DataKinds and GADT

“Build them precisely by hand”.

- ① `Support.CT0075DataKindCategory.hs`

Exercises.

- Implement the category **Terminal** adapting what has just been done at Point 1
(`Support.CT0050CatTerminal.hs` is a possible solution)
- Implement the category **Arrow** adapting what has just been done at Point 1
(`Support.CT0050CatZeroOne.hs` is a possible solution))

- `Support.CT0075ControlCategory.hs`

The “category” Hask

Category	Haskell
Identity	<code>id :: a -> a</code>
Object	Type (<code>Int</code> , <code>Bool</code> , <code>a</code>)
Morphism	Function (<code>a -> b</code>)
Composition	<code>(.) :: (b -> c) -> (a -> b) -> a -> c</code>